MIL-P-40619A
9 December 1968
SUPERSEDING
MIL-P-40619
3 April 1962
(See 6.3 and 6.4)

MILITARY SPECIFICATION

PLASTIC MATERIAL, CELLULAR, POLYSTYRENE

(FOR BUOYANCY APPLICATIONS)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers cellular polystyrene of two classes and two grades, intended primarily for buoyance applications where a rigid material is required.

1.2 Classification. Cellular (foamed) polystyrene shall be furnished in the following classes and grades, as specified (see 6.2).

Class 1 - Non-self-extinguishing.
Grade A - Low density.
Grade B - High density.
Class 2 - Self-extinguishing.
Grade A - Low density.
Grade B - High density.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
C-C-91 - Candles, Illuminating.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
C 203 - Test for Breaking Load and Calculated Flexural Strength of Performed Block-Type Thermal Insulation.
D 471 - Test for change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids.
D 1621 - Test for Compressive Strength of Rigid Cellular Plastics.
D 1622 - Test for Apparent Density of Rigid Cellular Materials.

FSC 9330

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) (CONT'D)
D 1667 - Test for Sponge Made from Closed Cell Poly (Vinyl Chloride), or Copolymers Thereof.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Cellular polystyrene shall be an essentially closed cell rigid form and shall be formed by the expansion and fusion of polystyrene beads in a mold or by mixing polystyrene under heat and pressure with a vaporizable substance with subquent expansion during an extrusion process.

3.2 Uniformity. The foamed material shall be of uniform texture and density throughout its mass. Repairs of any kind to the expanded material shall not be permitted.

3.3 Color. The color of class 1 material shall be white or the natural color of the product. Class 2 material shall be of a solid color other than white or marked "SELF EXTINGUISHING" in such a manner as prescribed by the procuring activity (see 6.1). Marking shall not degrade the physical properties of the subject material.

3.4 Odor. The material shall be free of objectionable odor.

3.5 Form. The material shall be in the form of boards, rough planks or billets. Unless otherwise specified (see 6.1), boards shall be furnished. Edges of boards shall be square within 1/8 inch per foot.

3.6 Dimensions and tolerances. Length, width, and thickness dimensions shall be as specified in table I (see 6.1).

Table I - Tolerances (for boards only).

Dimensions	Tolerances Inches
Length	
up to 48 inches	± 1/4
above 48 inches	+2, -1/2
Width	± 1/16
Thickness	± 1/16

3.7 Physical properties. The cellular polystyrene shall meet the requirements listed in table II when tested as specified therein.

Table II - Physical properties.

Property	Test method reference	Requirement Classes 1 and 2	
		Grade A	Grade B
Density, pounds per cubic foot	4.4.2		
Maximum		2.50	6.50
Minimum		1.50	6.00
Compressive strength, psi, minimum	4.4.3	18.0	100.0
Flexural strength, psi minimum	4.4.9	20.0	100.0
Water absorption, pounds per square foot of skinless surface, maximum	4.4.4	0.12	0.10
Volume change on heat aging, percent, maximum	4.4.5	3.5	3.5

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Table II - Physical properties. (Continued)

Property	Test method reference	Requirement Classes 1 and 2 Grade A	Grade B
Flame retardance (class 2 only)	4.4.6	Self extinguishing	Self extinguishing
Oil resistance	4.4.7	No softening, visible volume change, or deterioration	
Resistance against crushing (falling ball)	4.4.8		
Depth of depression, inches, maximum		0.24	0.21
Cracking beyond depression		No cracks	No cracks

3.8 Adhesives. If required by the procuring activity (see 6.1), the contractor shall recommend water resistant, air drying adhesives for joining cellular polystyrene. The resulting jointed foam shall meet all the applicable requirements of this specification.

3.9 Identification marking. One face of the foamed polystyrene planks, boards, or billets shall be marked permanently and legibly at recurrent intervals of not more than 2 feet with the manufacturer's identification, specification number, class, and grade. Federal Stock Catalog number, if any, may be used in lieu of specification number.

3.10 Workmanship. Workmanship shall be uniform and shall conform to best commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purpose of sampling, examinations, and tests, a lot shall consist of all the material of the same class and grade produced under essentially the same conditions, and offered for delivery at one time.

4.2.2 Unit for examination of preparation for delivery. The sample unit of examination of material (see 4.3.1) shall be one full length of board, plank, or billet. The sample unit for examination of preparation for delivery (see 4.3.2) shall be one pack, fully prepared for delivery.

4.2.3 Sampling for examination. For the examinations of 4.3.1 and 4.3.2 random samples shall be taken in accordance with MIL-STD-105.

4.2.4 Sampling for tests. Representative samples for testing shall be taken at random from those which passed the examination of 4.3.1. These samples shall be used for the tests of 4.3.5 and 4.3.6.

4.3 Quality conformance examination and tests.

4.3.1 Examination for defects of material. Samples shall be taken from each lot at inspection Level II and shall be examined at the acceptable quality level (AQL) 4.0 percent defective for conformance with the requirements for uniformity (see 3.2), color (see 3.3), odor (see 3.4), form (see 3.5), dimensions and tolerances (see 3.6 and 6.1), and workmanship (see 3.10). Defects are listed in table III.

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Table III - Defects of materials.

Examine	Defects
Uniformity and appearance	Major Texture not uniform; surface non continuous over entire area; repairs to foam. Minor Surface soiled, or marred.
Color	Major Color of class 1 not white or natural. Class 2 not colored or not distinguished from class 1.
Odor	Major Objectionable.
Form	Major Not as specified; edges not square within 1/8 inch per foot (for boards only).
Dimensions and tolerances	Major Dimensions not as specified, exceeding tolerances.
Identification marking	Major Not marked as specified. Not marked either permanently, legibly, or at recurrent intervals. Minor Marked recurrently at intervals greater than 2 feet.
Workmanship	Major Not uniform; largest dimension of voids greater than 1/8 inch; presence of resin accumulations; presence of foreign inclusions; crushed surface areas larger than 10 percent of board surface and deeper than 5 percent of original thickness; tears, cracks extending a distance larger than 1/2 inch from edges; defects detracting from serviceability; repairs.
Workmanship	Minor Tears or cracks less than 1/2 inch from edges; crushed surface areas not larger than 10 percent of board surface and not deeper than 5 percent of original thickness; edges and corners rounded (except for billets).

4.3.2 Examination for defects in preparation for delivery. Samples shall be taken from each lot at inspection level S-2 and shall be examined at the acceptable quality level (AQL) 4.0 percent defective for conformance with Section 5 of this specification. Defects are listed in Table IV.

Table IV - Defects in preparation for delivery.

Examine	Defects
Packing	Major Not specified level; shipping container not as specified; number of boards, planks or billets per pack not as specified, if required (see 6.1). Minor Method of closure not as specified.

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Table IV - Defects in preparation for delivery. (Continued)

Examine	Defects
Marking	Major Not indicating class and grade; illegible, incorrect; not conforming to carrier regulations. Minor Incomplete, not in accordance with contract requirements; special markings missing.

4.3.3 Definition of defects. A major defect is likely to prevent or to reduce materially the usability of the unit of product for its intended purpose. A minor defect is not likely to reduce materially the usability of the unit of product for its intended purpose. Total defects are the sum of major and minor defects. For the purpose of computing total defects two minor defects shall be counted as one major defect.

4.3.4 Rejection. Any sample unit of 4.2.3 found defective shall be rejected and if the number of total defects exceeds the applicable acceptance number, the entire lot represented by the sample unit shall be rejected.

4.3.5 Production check tests. Production check tests shall be conducted on samples taken in accordance with 4.2.4 from or representing the first lot of material offered for delivery and from one out of every five lots thereafter, but not later than one year after delivery of the first lot, whichever is the shorter period. All the tests specified in 4.4 shall be performed.

4.3.6 Quality conformance tests. Quality conformance tests shall be conducted on samples taken in accordance with 4.2.4 from all lots on which production check tests were not conducted. The tests for density (see 4.4.2), water absorption (see 4.4.4), and flame retardance (see 4.4.6) shall be performed, the latter for Class 2 only.

4.3.7 Action in case of nonconformance. If any of the samples in the production check tests or quality conformance tests are found not to be in conformance with the requirements of this specification, the lot which it represents shall be rejected. Futhermore, additional samples shall be taken or provided from each subsequent lot and shall be subjected to the test or tests wherein the failure occurred. Each lot shall then be accepted only after satisfactory results are obtained on the test or tests by all samples taken or provided to represent the lot. This additional testing shall be discontinued after four successive lots have passed the test or tests.

4.4 Test procedures.

4.4.1 General conditions. The specimens shall be conditioned and tested at 73.4° ± 9° Fahrenheit (F) (23° ± 5° Centigrade (C.)) and 50 ± 4 percent relative humidity. In cases of dispute, the conditions 73.4° ± 1.8°F. (23° ± 1°C.) and 50 ± 2 percent relative humidity shall apply. Unless otherwise specified, not less than 2 specimens per sample unit shall be tested.

4.4.2 Density. The density shall be determined by ASTM D 1622.

4.4.3 Compressive strength. The compressive strength shall be determined by procedure A of ASTM D 1621. The pressure shall be applied until failure or until the specimen has been compressed 5 percent of the original thickness, whichever occurs first.

4.4.4 Water absorption. The water absorption shall be determined by ASTM D 1667.

4.4.5 Volume change on heat aging. The volume before and after heat aging shall be determined by the displacement method. The confining medium shall be a free flowing, non dusting solid of small particle size. Clean flax or rape seed is suitable. The specimens shall be of suitable size to fit into a 200 milliliter (ml.) cylinder with 1 ml. graduations and shall allow covering the specimen by the confining medium within the graduated section of the cylinder. The specimen shall be placed in the cylinder, surrounded by the measuring medium and the total volume shall be determined. The difference between this total volume and that of the measuring medium only is the volume of the specimen. The specimen shall then be heated in an oven for 48 ± 1/4 hours at 140°F. (60°C.). After conditioning, the volume of the specimen shall be determined as described above. The volume

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change after aging expressed in percent of the initial volume shall be calculated using the following expression:

$$V = \frac{v_i - v_a}{v_i} \times 100$$

Where : V = Volume change in percent of initial
v_i = Initial volume
v_a = Volume after aging

4.4.6 Flame retardance.

4.4.6.1 Specimens. The specimens shall be one-inch wide and 1/4 inch thick with a tolerance of + 1/16 inch. The length shall be 12 inches with a tolerance of ± 1/16 inch.

4.4.6.2 Test equipment. The equipment shall include a means of supporting and positioning the specimen. A laboratory ring stand and burette clamp are suitable. A candle conforming to C-C-91 and a timing device permitting measurement of time intervals of 0.2 second are required.

4.4.6.3 Test procedure. The test shall be conducted in a location free from drafts. The specimen shall be clamped at the upper end with the length dimension positioned downward at a 45° angle from the horizontal and the width dimension vertical. The skinless 1/4 inch face shall be in a downward position facing the flame. The candle (see 4.4.6.2) shall be prepared for use by allowing it to burn until the conical top is completely consumed and until the wick has assumed a bent shape. The wick shall not be trimmed. The flame shall be extinguished and the clamp holding the specimen adjusted to provide a vertical clearance of 0.25 inch between the highest point of the bent wick and the lowest point on the specimen. The candle shall be removed, lighted and permitted to burn until molten wax fills the crater forming the top of the candle. The burning candle shall then be replaced under the specimen. If the burning portion of the specimen melts or shrinks from the flame, the candle shall be moved along the 45 degree angle so as to keep the specimen in the flame. At the end of 15 ± 0.2 seconds the candle shall be removed. The burning plus glowing times of the specimen after withdrawal of the candle shall be measured. Ten specimens shall be used in one test. The average of the burning plus glowing times shall not be longer than ten seconds after removal of the flame in order for the material to be rated self extinguishing by this test. If any specimen is completely consumed during the 15 second exposure period, a new specimen shall be substituted and if that specimen is also consumed, the material shall be rated "non-self-extinguishing". If more than one of the ten specimens used in the test is completely consumed, the material shall be rated "non-self-extinguishing".

4.4.7 Oil resistance. Specimens of suitable size shall be immersed for 70 + 1/2 hours in No. 2 oil of ASTM D 471. The specimen shall then be removed, blotted gently with filter paper and compared with an untreated specimen of similar size for visible volume change, softening or deterioration.

4.4.8 Resistance against crushing.

4.4.8.1 Specimens. Specimens shall be 4 by 4 by 1 inches. If the exterior of the finished article is intended to have a skin (see 6.1) one of the 4 by 4 inch surfaces of the specimens shall be completely covered by an integral skin. Otherwise the presence of a skin on the test face of the specimen shall be optional.

4.4.8.2 Test procedure. A specimen with surface skin up, if present shall be placed on a hard flat surface, such as metal or concrete. A steel ball, 2.14 inches in diameter weighing 1.44 pounds, shall be dropped once vertically from a height of four feet, measured from the hard surface, on the center of the specimen. The ball shall be removed and after a recovery period of 15 minutes the depth of the formed depression shall be measured to the nearest 0.01 inch.

4.4.9 Flexural strength. The flexural strength shall be determined by ASTM C 203.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

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5.1 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into equipment and lose their separate identity when the equipment is shipped.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Packing shall be in fiberboard boxes conforming to PPP-B-636, Type CF, class weather-resistant, grade V3c, or PPP-B-640, class 2, style A. All center and edge seams and the manufacturer's joint of the boxes shall be waterproofed in accordance with the applicable box specification or appendix thereto. Box closure and reinforcing shall be in accordance with the appendix to the applicable box specification except reinforcing shall be by pressure sensitive, reinforced filament tape. The gross weight of shipping containers shall not exceed the weight limitations of the applicable box specification. Dimensional limits shall be waived.

5.2.2 Level B. Packing shall be in fiberboard boxes conforming to PPP-B-636, class domestic or PPP-B-640, class 1, except that dimensional limits shall be waived. Box closures shall be as specified in the applicable box specification or appendix thereto. Gross weight of shipping containers shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. Packing shall be accomplished in a manner which will insure acceptance by a carrier at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early material use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. Shipment marking information shall be provided on shipping containers in accordance with suppliers commerical practice provided such meets the requirements of MIL-STD-129. The information shall include nomenclature, Federal Stock number or manufacturer's part number, lot number, date (month and year) of manufacture, size, contract or order number, contractor's name, and designation. Special markings shall be provided if specified (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and grade required (see 1.2).
(c) Manner in which "SELF EXTINGUISHING" shall be marked (see 3.3).
(d) Form, if other than board (see 3.5).
(e) Dimensions (see 3.6).
(f) Recommendations for adhesives, if required (see 3.8).
(g) If exterior surface of finished article shall have a skin (see 4.4.8.1).
(h) Number of items per pack, if required (see 4.3.2).
(i) Level of packing (see 5.2).
(j) Special markings, if required (see 5.3).

6.2 Flame retardance. The term "self-extinguishing" denotes solely that burning class 2 material will self-extinguish itself, when an external fuel source is removed. Class 2 material represents a lesser fire hazard than does Class 1 material, but the term "self-extinguishing" should not be considered a fire hazard classification.

6.3 Supersession data. This specification supersedes, classes 1 and 2, grades A and B of Type II of MIL-P-40619. Type I classes 1 and 2, grades A and B of MIL-P-40619 are included in Federal Specification HH-I-524a, dated May 23. 1966.

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6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Navy - SH
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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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